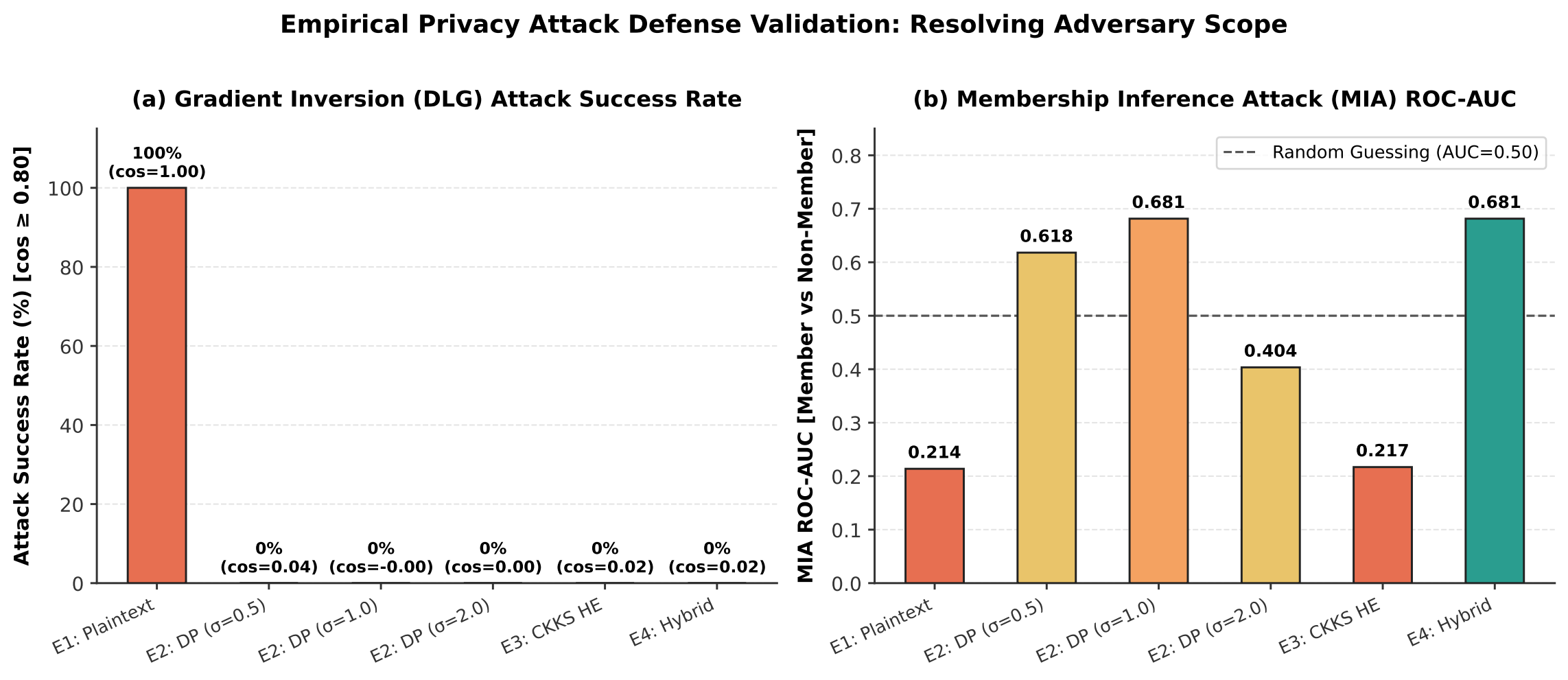
Empirical Privacy Attack Defense Validation: Resolving Adversary Scope
(a) Gradient Inversion (DLG) Attack Success Rate
Attack Success Rate (%) [cos ≥ 0.80]
100%
(cos=1.00)
0%
(cos=0.04)
0%
(cos=-0.00)
0%
(cos=0.00)
0%
(cos=0.02)
0%
(cos=0.02)
E1: Plaintext
E2: DP (σ=0.5)
E2: DP (σ=1.0)
E2: DP (σ=2.0)
E3: CKKS HE
E4: Hybrid
(b) Membership Inference Attack (MIA) ROC-AUC
MIA ROC-AUC [Member vs Non-Member]
Random Guessing (AUC=0.50)
0.214
0.618
0.681
0.404
0.217
0.681